

CREDIT RISK MODEL FAIRNESS AUDIT

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Scope: Logistic Regression Model for credit risk, evaluated under EU AI Act (Annex III, 5(b) – Creditworthiness Assessment)

Protected Attribute: Age (< 25)

EXECUTIVE SUMMARY

This audit evaluates a logistic regression model for credit risk under the EU AI Act's high-risk classification. The protected attribute is age, with applicants under 25 designated as the protected group.

Key Findings: The baseline model exhibited disparate impact against the protected group, approving credit applications at a 40% lower rate than the reference group (age ≥ 25). This corresponds to a Disparate Impact (DI) ratio of 0.60, violating the 80% rule—a widely adopted benchmark in financial fairness auditing.

Remediation: A bounded grid search over decision threshold and logit shift was applied. The selected configuration achieved a DI ratio of 0.84 on the validation set, which falls within the 0.80–1.25 compliance band, and reduced the estimated financial loss by €5,240 (from €57,663 to €52,423).

Final Evaluation: The remediated model was evaluated on a held-out test set. The final DI ratio is 1.12, which remains within the compliance band and stable across FN:FP cost ratios from 2:1 to 6:1 (including the estimated 3.4:1 ratio). However, equalized odds fails at lower ratios (1:1 and 1.5:1). While the validation set FPR ratio (1.02) meets the 0.80–1.25 band, the test set FPR ratio (0.69) falls below it. Wide confidence intervals (DI 95% CI: 0.58–1.85) indicate high statistical uncertainty due to the limited test sample (n=200). This finding should be treated as directional, not conclusive, until validated on a larger cohort.

Secondary Findings:

1. **Manual Review Disparity:** Applicants under 25 are routed to manual review at $\sim 1.38\times$ the rate of older applicants, suggesting age-related disparity extends beyond automated decisions and warrants operational review.
2. **Proxy Discrimination Risk:** Remaining features predict foreign worker status with AUC = 0.738, indicating potential indirect bias that persists despite protected attribute exclusion.
3. **Foreign Worker Assessment Limited:** Insufficient native worker representation (n=37 full dataset, n=6 test set) prevents statistically reliable DI evaluation for this group.
4. **Sex Attribute Note:** DI ratio for female applicants is 1.15 (within band). However, the dataset's `personal_status` field conflates sex with marital status, and the single-female category contains too few observations to yield reliable estimates, limiting interpretability.

Statistical Limitations: The test set contains only 200 samples. Bootstrapped confidence intervals are wide, meaning the compliant point estimate should be treated as directional, not conclusive, until validated on a larger, representative dataset.

Recommendation: This audit does not conclude full regulatory readiness. While DI compliance for age is achieved at the selected operating point, the following require resolution before production deployment:

- Equalized odds failure on the test set
- Unresolved disparity in manual review escalation
- Proxy discrimination risk regarding nationality
- Statistical uncertainty due to sample size

Before deployment, the audit recommends: (1) validating these findings on a larger, representative dataset; and (2) adopting an in-processing fairness-constrained approach (e.g., Fairlearn's ExponentiatedGradient) to embed fairness during training and remove dependency on protected attributes at inference time.

Full technical documentation, reproducible audit code, and JSON audit trail are available in the project repository upon request.